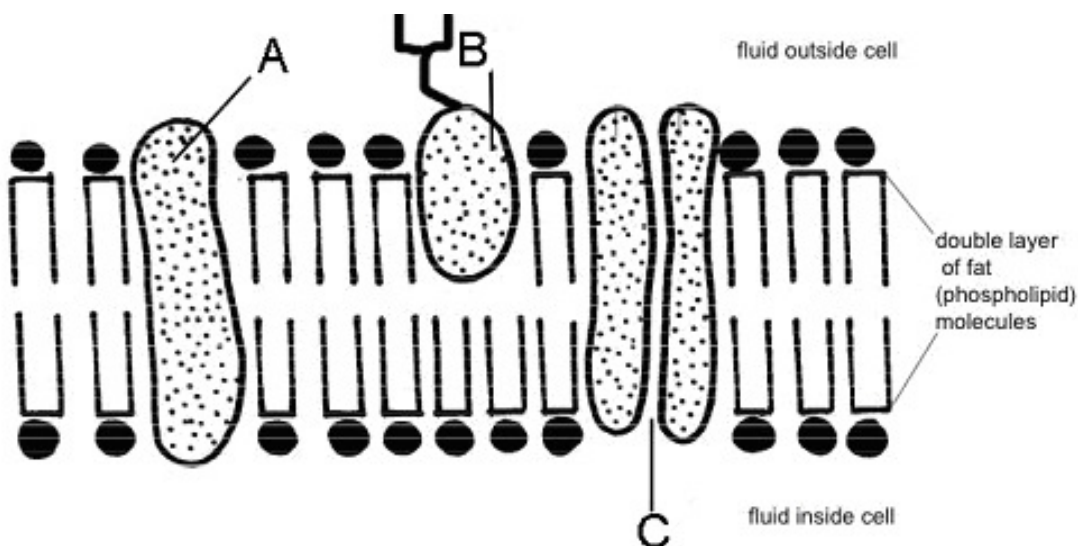


- 1 Some cellular organelles are bound by a single membrane, while other organelles have two membrane envelopes around them. Which of the following is correct?

	single membrane		two membranes	
A	tonoplast	lysosome	nucleus	chloroplast
B	chloroplast	lysosome	nucleus	tonoplast
C	nucleus	lysosome	chloroplast	tonoplast
D	tonoplast	chloroplast	nucleus	lysosome

- 2 The diagram shows a simplified model for the plasma membrane of an eukaryotic cell.

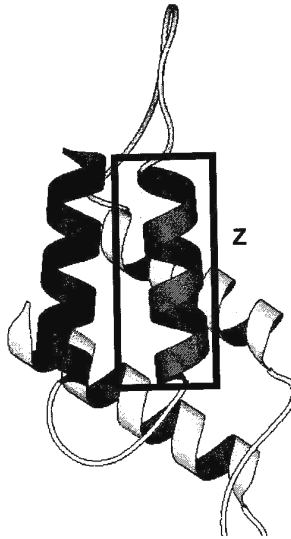


(adapted from Wikipedia)

Which option shows the most likely scenario for the three labeled proteins?

	protein A	protein B	protein C
A	glycoprotein 120	calcium channel	G protein coupled receptor
B	tyrosine kinase receptor	glycoprotein 120	chloride channel
C	glycoprotein 41	G-protein coupled receptor	sodium channel
D	cytoskeleton	glycoprotein 41	potassium channel

- 3 Which one of the following statements is **INCORRECT**?
- A** There are 3 polypeptide chains in the collagen helix, and only one in the α - helix
 - B** Proline is a major component of collagen but not of the α -helix.
 - C** Every third residue in the α -helix must be glycine, but there is no sequence regularity in collagen
 - D** Some hydrogen bonding exists between different chains in collagen, but hydrogen bonding occurs within the same chain in the α -helix.
- 4 The figure below show the structure of a protein molecule. .

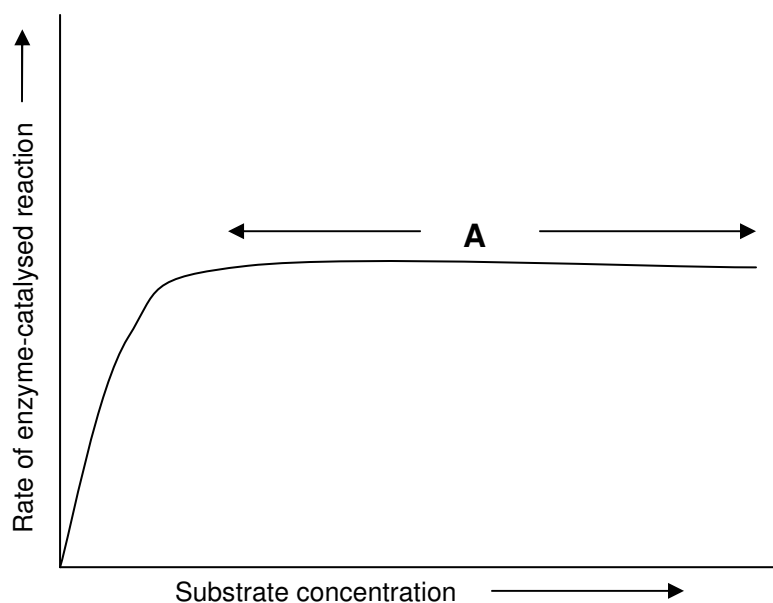


Which of the following correctly describes the structure shown in Box **Z**?

	Number of amino acids present	Types of bond(s) present
A	Cannot be determined	Hydrogen bonds only
B	Cannot be determined	Hydrogen bonds and hydrophobic bonds
C	Approximately 11 amino acids	Hydrogen bonds only
D	Approximately 7 amino acids	Hydrogen bonds and hydrophobic bonds

- 5 The graph shows the results of an experiment to determine the effect of varying substrate concentrations on the rate of an enzyme-catalysed reaction.

The experiment was conducted at the optimum pH for the enzyme and in the presence of an inhibitor.



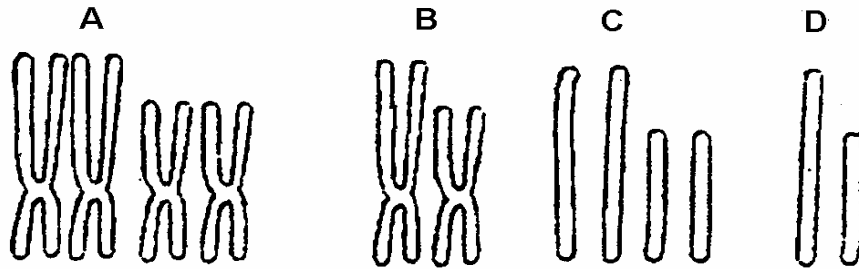
The rate of the enzyme-catalysed reaction in the section labeled **A** on the graph would be higher if

- A** the substrate concentration was increased.
 - B** the pH was increased.
 - C** the enzyme concentration was decreased.
 - D** the inhibitor concentration was decreased.
- 6 A bacterial amylase was incubated with starch at 20°C and pH 7. When a sample was removed after 5 minutes and mixed with iodine reagent the mixture was light brown colour. The most likely explanation of this observation is that
- A** an inhibitor of the amylase was present
 - B** the pH was too high to allow the reaction to proceed quickly
 - C** the starch in the reaction mixture had been converted to maltose
 - D** bacterial amylase does not catalyse the digestion of starch

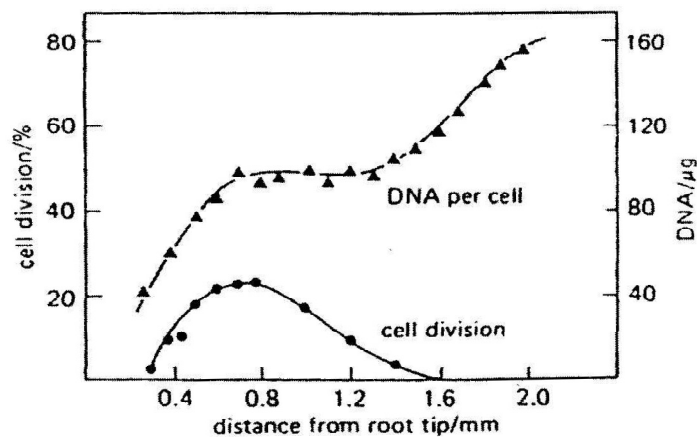
- 7 The diagram shows the chromosome of a cell just before the metaphase stage of mitosis.



What will be the appearance of the chromosomes at telophase?



- 8 The graph below shows the changes in DNA content and rate of division in cells at varying distances from the tip of an onion root.



Which one of the following can be determined from the graph?

- A Cells nearest the tip are dividing by meiosis.
- B DNA quantity varies inversely with the cell division rate.
- C DNA synthesis precedes cell division
- D The cells are becoming polyploidy.

- 9** Identify the correct statements regarding the replication of DNA
- 1 Only 1 strand of DNA is replicated
 - 2 It occurs in S phase of interphase in both mitosis and meiosis
 - 3 It occurs after the completion of telophase in meiosis
 - 4 DNA polymerase III is required (for bacteria)
- A** 1 and 3 only
- B** 2 and 4 only
- C** 2 and 3 only
- D** 2 only
- 10** The following statements describe how replication of DNA in bacteria would proceed if various **enzymes** are absent
- 1 Without enzyme A, double-stranded DNA separated into single strands may re-anneal.
 - 2 Without enzyme B, DNA remains as a double helix, unless excessive heat close to 100 °C is provided to it.
 - 3 Without enzyme C, short Okazaki fragments will not be joined in the lagging strand.
 - 4 Without enzyme D, wrong DNA nucleotides added to a newly synthesized strand would not be replaced.

What are enzymes A, B, C and D?

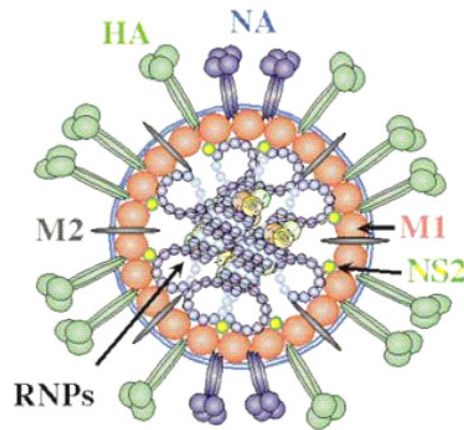
	enzyme A	enzyme B	enzyme C	enzyme D
A	helicase	DNA polymerase II	ligase	DNA polymerase III
B	ligase	DNA polymerase III	helicase	DNA polymerase II
C	DNA polymerase III	helicase	ligase	DNA polymerase II
D	DNA polymerase II	DNA polymerase III	helicase	ligase

- 11** A particular sequence in one of the DNA strands (strand that is complementary to the strand used as a template for transcription) reads CAT-CGG.

Which one of the following shows the corresponding sequence for the anti-codons?

- A** GTA-GCC
 - B** GGC-AUG
 - C** CAU-CGG
 - D** GUA-GCC
- 12** Double stranded viruses can reproduce via the lytic and lysogenic cycle. Which of the following is characteristic of the lytic cycle?
- A** Viral DNA is incorporated into host chromosomes.
 - B** Viral DNA is incorporated into the host chromosome.
 - C** The host-virus relationship usually lasts for generations.
 - D** A large number of phages are released at a time.

- 13 Refer to the diagram below showing the structure of virus X



Virus X is a/ an

- A Bacteriophage lambda
 - B Orthomyxovirus
 - C Herpes virus
 - D Human immunodeficiency virus
- 14 A gene transfer occurred between two strains of bacteria. It was observed that cell-cell contact is **NOT** required, and a filter that removes viruses eliminates transfer. This would mean that the transfer occurred by
- A conjugation.
 - B transformation.
 - C transposition.
 - D transduction.

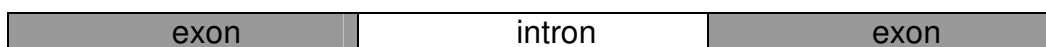
- 15 Which of the following scenario has the highest risk of cancer?

	proto-oncogene	tumour suppressor gene
A	gain of function in one allele	loss of function in both alleles
B	gain of function in both alleles	loss of function in one allele
C	loss of function in both alleles	gain of function in one allele
D	loss of function in one allele	gain of function in one allele

- 16 Gene amplification is a natural process in a cell in which a particular gene is replicated so that many more copies of the gene are available for transcription immediately.

Which of the following is an example of a gene that is most likely to be amplified?

- A genes coding for haemoglobin breakdown in a liver cell
 - B genes coding for protein channels / pumps in neurons
 - C genes coding for rRNA and tRNA in an early embryo
 - D genes coding for digestive enzymes in a gut cell
- 17 The diagram shows a gene in a human cell



In one form of the blood condition thalassaemia, there is a mutation in an intron of the haemoglobin gene.

Which stage of gene expression would this mutation directly affect?

- A activation of the gene by a steroid hormone
- B attachment of mRNA to a ribosome
- C binding of RNA polymerase to DNA
- D splicing of pre-mRNA

18 Four different genes are expressed in different ways.

Gene 1 undergoes tissue-specific patterns of alternative splicing

Gene 2 is part of a group of structural genes controlled by the same regulatory sequences

Gene 3 is in some circumstances subject to methylation

Gene 4 codes for a repressor protein which acts at an operator site close by

Which of the above genes are of eukaryotic origin?

- A** 1 only
- B** 1 and 3 only
- C** 2 and 3 only
- D** 2, 3 and 4 only

19 Which of the listed features are characteristics of an eukaryotic genome?

- 1 circular DNA
- 2 DNA associated with histone
- 3 DNA complexed to form nucleosome
- 4 DNA may be transferred to new cells by mitosis

- A** 1 only
- B** 1 and 3 only
- C** 2 and 3 only
- D** 2, 3 and 4 only

- 20** A farmer performed the chi-square test (also termed the goodness-of-fit test) on the observed phenotypic ratio in the F₂ generation of a dihybrid cross, expecting a ratio of 9:3:3:1. The chi-square value obtained in his calculation was 6.66.

Using a portion of the chi-square table below to help you, which is the most likely conclusion for the farmer?

Degrees of freedom	Probability			
	0.95	0.50	0.10	0.05
1	0.004	0.46	2.71	3.84
2	0.10	1.39	4.60	5.99
3	0.35	2.37	6.25	7.82
4	0.71	3.36	7.78	9.49

- A** The calculated chi-square value is higher than the critical chi-square value; the null hypothesis is accepted
- B** The calculated chi-square value is higher than the critical chi-square value; the null hypothesis is rejected
- C** The calculated chi-square value is lower than the critical chi-square value; the null hypothesis is accepted
- D** The calculated chi-square value is lower than the critical chi-square value; the null hypothesis is rejected
- 21** A color blind man (who is otherwise normal) marries a woman who has normal vision but is hemophilic. Given that both these traits are X-linked and recessive, which of the following can be expected from their offspring

	sons		daughters	
A	50% hemophilic, normal vision	50% non-hemophilic, normal vision	100% hemophilic and color blind	
B	100% normal vision and hemophilic		100% carriers for color blindness and hemophilic	
C	100% carriers for hemophilia and color blindness		100% carriers for color blindness	
D	50% carriers for hemophilia	50% normal	50% carriers for color blindness	50% carriers for hemophilia

- 22** A man with **O** blood type married a woman with **AB** blood type and they have four children in the family. The first has **A** blood type, the second has **B** blood type, the third has **AB** blood type and the fourth has **O** blood type.

Which of the following conclusions is possible from the information given?

- A** The first two children are adopted as they do not share the blood type of their parents
 - B** The third and fourth children are probably adopted as the parents cannot produce children with these phenotypes
 - C** Only the third child belonged to the parents and the others are probably adopted as the **A** and **B** allele are codominant
 - D** It is possible that all four children are the offspring of the pair of parents because of independent assortment of chromosomes
- 23** Six tubes containing preparations from animal tissue were set up as shown below.

<i>Tube</i>	<i>Contents</i>
1	Glucose + homogenised cells
2	Glucose + mitochondria
3	Glucose + cytoplasm lacking organelles
4	Pyruvic acid + homogenised cells
5	Pyruvic acid + mitochondria
6	Pyruvic acid + cytoplasm lacking organelles

After incubation, in which three tubes would carbon dioxide be produced?

- A** 1, 2 and 3
- B** 1, 4 and 5
- C** 3, 4 and 6
- D** 4, 5 and 6

24 Choose the **INCORRECT** statement concerning the chemiosmotic mechanism of the coupling of substrate oxidation to the phosphorylation of ADP.

- A** The main protein complexes of the electron transport chain in mitochondria pump protons out into the intermembrane space
- B** A central role is played by a proton gradient that exists across the inner mitochondrial membrane.
- C** The pH inside the mitochondrial matrix is less than that in the cell cytoplasm
- D** The electrochemical 'force' which drives the synthesis of ATP is called the proton-motive force.

25 The table shows the amount of carbon dioxide taken up by a plant in the light and the amount of carbon dioxide produced in the dark, at different temperatures.

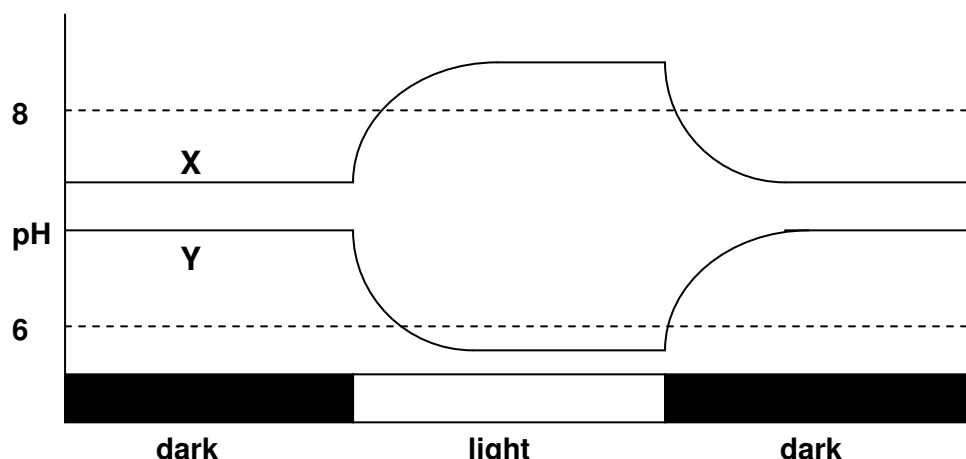
Temperature °C	5	15	30
Net uptake of CO ₂ in light (mg g ⁻¹ h ⁻¹)	1.1	2.8	2.0
Release of CO ₂ in dark (mg g ⁻¹ h ⁻¹)	0.2	0.8	2.4

Assuming that the rate of respiration of this plant is the same in light and dark, decide which of the following statements are **TRUE**.

(i)	Temperature has no effect on the rate of carbon dioxide uptake in the light.
(ii)	The true uptake of carbon dioxide in the light at 15 °C is 3.6mg g ⁻¹ h ⁻¹
(iii)	The true uptake of carbon dioxide in the light at 15 °C is 2.0mg g ⁻¹ h ⁻¹
(iv)	The rate of photosynthesis exceeds the rate of respiration at all temperatures.
(v)	The true uptake of carbon dioxide in the light at 15 °C is 2.8mg g ⁻¹ h ⁻¹

- A** i & iii
- B** ii & v
- C** iii & iv
- D** ii & iv

- 26 The diagram below consists of two graphs showing the changes in pH of two structures **X** and **Y**, of a chloroplast in the absence and presence of light. Which of the following correctly identifies structures **X** and **Y**?



	X	Y	Reason
A	Thylakoid space	Stroma	Protons are pumped from thylakoid space into the stroma hence decreasing the latter's pH.
B	Stroma	Thylakoid space	Protons are pumped from stroma into the thylakoid space hence decreasing the latter's pH.
C	Thylakoid membrane	Stroma	Protons are pumped from thylakoid membrane into the stroma hence decreasing the latter's pH.
D	stroma	Thylakoid membrane	Protons are pumped from stroma into the thylakoid membrane hence decreasing the latter's pH.

- 27 Which of the following correctly describes the different components of a homeostatic mechanism for glucose regulation?

	stimulus	receptor	control centre	signal	effector
A	High glucose	islets of Langerhans	liver	glucagon	liver; muscles
B	High glucose	islets of Langerhans	islets of Langerhans	insulin	liver, muscles
C	lack of glucose	islets of Langerhans	islets of Langerhans	insulin	glucose
D	lack of glucose	liver	islets of Langerhans	glucagon	glucose

- 28 Impulses travel more rapidly along nerves from the leg muscles of a mammal than that along leg muscles of an insect because in the mammal:

- A** there is a high concentration of sodium ions inside the axon
- B** the nerves contain myelinated fibres
- C** there is a potential difference across the axon membrane
- D** the area of a cross section of a fibre is greater.

- 29 Curare causes paralysis of striated muscle and is used during general anaesthesia.

When curare is applied to a nerve-muscle preparation, the muscle does not contract when the nerve is stimulated, but if the stimulus is applied directly to the muscle, contraction occurs.

When curare is applied to the nerve only, the muscle contracts when the nerve is stimulated.

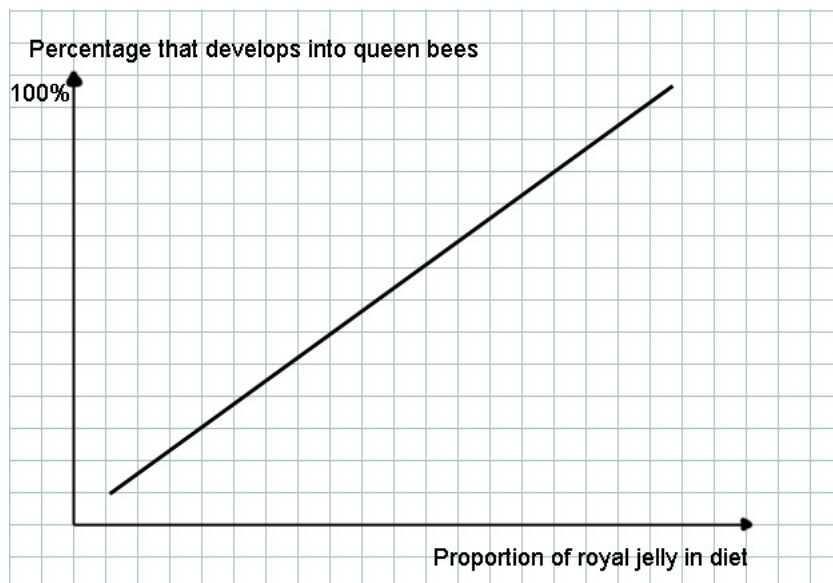
Which of the following statements correctly explains the effect of curare?

- A** It affects the formation of the enzyme cholinesterase
- B** It inhibits the development of an action potential in the nerve fibre.
- C** It prevents the depolarization of the post-synaptic membrane.
- D** It stimulates the formation of acetylcholine at the motor end plate.

- 30 Which of the following most likely shows the mechanism of cell signaling for a steroid hormone?
- A Bind to an intracellular receptor to enter nucleus
 - B Bind to a G-protein coupled receptor to activate G-protein
 - C Bind to receptor tyrosine kinase for the purpose of phosphorylation of other enzymes
 - D Bind to a ligand gated ion channel to allow the movement of ions across the membrane.
- 31 Which of the following is true of continuous and discontinuous variation?

	continuous variation	discontinuous variation
A	many intermediates, controlled by several genes	few intermediates, controlled by one or two genes
B	few intermediates, controlled by one or two genes	many intermediates, controlled by several genes
C	few intermediates, controlled by several genes	many intermediates, controlled by one or two genes
D	many intermediates, controlled by one or two genes	few intermediates, controlled by several genes

- 32** Analyse the following graph about worker honeybees fed on royal jelly and select the best conclusion based the information provided.



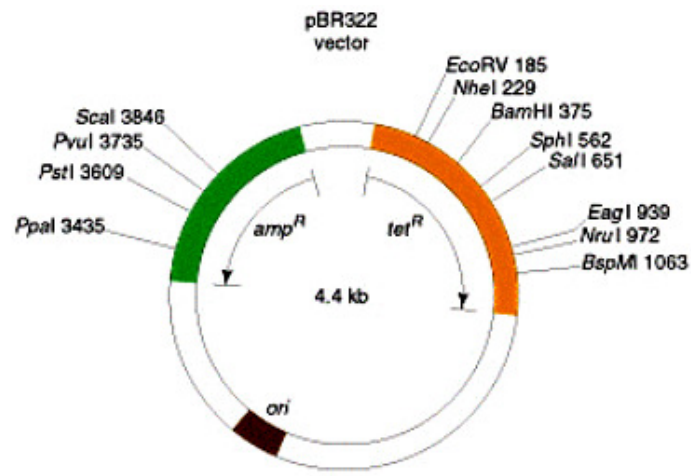
- A** Worker bees develop into queen bees only when fed exclusively on royal jelly.
 - B** Worker bees develop into queen bees over time.
 - C** Having a higher proportion of royal jelly in the diet causes genetic changes in honeybees.
 - D** The phenotype of the honeybee is dependent on diet.
- 33** A large population of equal numbers of dark and light mice was released into an area where owls are predators of mice. Because of predation, only 25% of these mice survived and selective killing by owls changed the proportions of dark to light mice from 1:1 to 4:1.

If mice produce an average of 8 offspring per litter and the pattern of predation remains the same, what is most likely to happen to the population of light mice?

- A** They disappear after one further generation.
- B** They disappear after two further generations.
- C** They remain at the level of one fifth of the population.
- D** They would be reduced to a constant but very low frequency.

- 34** Which of the following characteristics is undesirable in cloning vectors used in genetic engineering?
- A** Control their own replication.
 - B** High copy number.
 - C** Small in size.
 - D** Vulnerable at several sites to a restriction enzyme.
- 35** Restriction enzymes are molecular scissors used in genetic engineering to cut up DNA. In nature where are restriction enzymes found and what is their natural role?
- A** In viruses; to splice host cells DNA.
 - B** In bacteria; to defend against viral invasion.
 - C** In yeast cells, to defend against foreign DNA invasion.
 - D** In viruses; to defend itself against mutation.

- 36 pBR322 vector is used to clone a eukaryotic gene which has been digested by the restriction enzyme *Bam*HI.



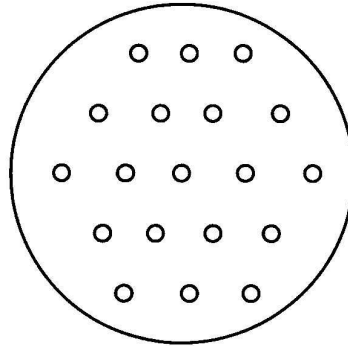
Following transformation, bacterial cells were grown in four different media, as shown below:

I	Nutrient broth plus ampicillin
II	Nutrient broth plus tetracycline
III	Nutrient broth plus ampicillin and tetracycline
IV	Nutrient broth without antibiotics

Which of the following media would bacterial cells that contain the recombinant plasmids grown in?

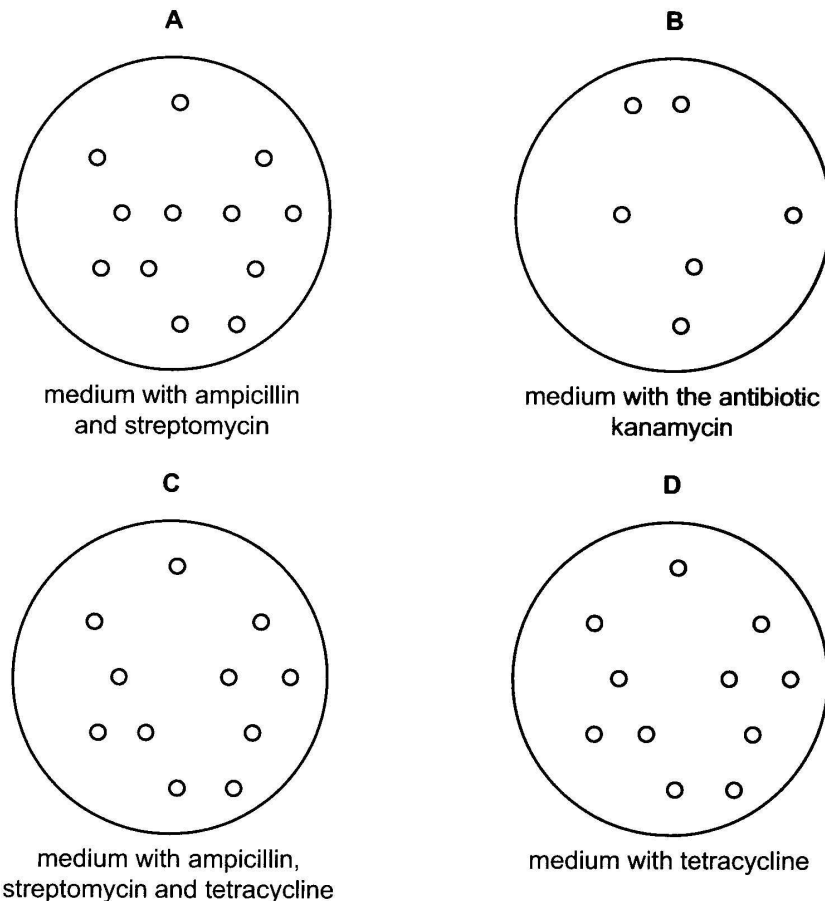
- A I and II
- B I and III
- C I and IV
- D IV only

- 37 A gene coding for the production of human gene product was inserted into plasmid with genes coding for resistance to antibiotics ampicillin, streptomycin and tetracycline. The plasmids were used to transform *E. coli* and the bacteria grown on a medium. The resulting master plate is shown in the diagram.



Transformed cells were selected by replica plating the bacteria colonies onto media containing various antibiotics.

Which plate contains the colony of bacteria into which the human gene has been successfully inserted?



- 38** Which statement correctly describes pluripotency?
- A** the ability to develop into a whole organism
 - B** the ability to develop into tissues from various cell types
 - C** the ability to develop into different cells of a particular tissue type
 - D** the ability to stay undifferentiated
- 39** Which process could produce a large number of genetically identical plants?
- A** pollen culture
 - B** embryo culture (plantlet)
 - C** meristematic tissue culture (protoplast)
 - D** anther culture
- 40** Which of the following is not an intended outcome for the use of DNA technology?
- A** to study the cause of genetic diseases so as to develop cures
 - B** to help solve the problems of malnutrition in third world countries
 - C** to tailor-make drugs (and dosage) according to individual needs
 - D** to improve the genome of the human species through gene therapy